Disfavored (95% C.L.), fixing $f_{e,S} = 1/3$

Today: IC MESE 11.4 yr flavor + spectrum

Today: IC HESE 7.5 yr + DR2 11.6 yr

Projected: combined detectors

$g_D \sum |U|^2 = 10^{-24}$

$g_D \sum |U|^2 = 10^{-16}$

$g_D \sum |U|^2 = 10^{-8}$

Coupled to ν_e

Coupled to ν_e

Coupled to ν_e

Coupled to ν_μ

Coupled to ν_μ

Coupled to ν_μ

Coupled to ν_τ

Coupled to ν_τ

Coupled to ν_τ

Kinetic mixing, $\log_{10} \epsilon'$

0

−10

−20

−30

−40

−50

−30 −25 −20 −15 −10 −5 0

Mediator mass, $\log_{10}(M_{Z_D}/\mathrm{eV})$